



PUBLIC COURSEWORK DOCUMENTATION

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2. INTRO / BACKGROUND

PH ecommerce reviews are messy. Code-switching breaks standard NLP models. Critical sentiment payloads inside emojis are completely ignored.

Sellers need instant, accurate feedback. Delayed responses to inquiries hurt metrics.

Develop an NLP-powered system that provides instant assistance and transparent sentiment classification for local dialects.

3. PROBLEM STATEMENT

- How can we accurately classify code-mixed Tagalog/Bisaya/English datasets?
- How do we extract mathematical sentiment from non-verbal emojis?
- How effective is the developed system in explaining its predictions to the user?

4. OBJECTIVES

Develop an NLP-integrated system specifically optimized for Philippine eCommerce data.

- Design the NLP system architecture.
- Implement emoji-aware NLP techniques.
- Evaluate classification performance.
- Test system usability and XAI response times.

5. SCOPE & LIMITATIONS

- **Languages:** English, Tagalog, Bisaya
- **Features:** Sentiment classification, Heuristic Tagging, Feature Impact extraction.
- **Target Users:** eCommerce sellers, Data Analysts.

LIMITATIONS

- Synthetic data used for training may contain template artifacts.
- Evaluation constrained to real-only test splits.
- No Deep Learning architectures (Transformers/BERT) utilized.

6. SYSTEM ARCHITECTURE



7. DATASET & PREPROCESSING

- **Source:** Shopee, Lazada, Play Store
- **Records:** 14,901 rows
- **Features:** Text, Sentiment, Language

- Tokenization (N-Grams)

- Emoji Translation:

```
{  
  "raw": "Ganda! 🥰",  
  "clean": "ganda EMOJI_smiling_face_with_heart_eyes"  
}
```

8. NLP TECHNIQUES

01

Converts text to mathematical vectors. Up-weights rare, domain-specific words (e.g. "sira") and down-weights common stop-words.

02

Used with `class_weight='balanced'`. Selected for linear interpretability. Allows us to dynamically extract exact coefficients for XAI visualization.

03

Custom algorithm to detect code-switching by intersecting token sets with language-specific markers.

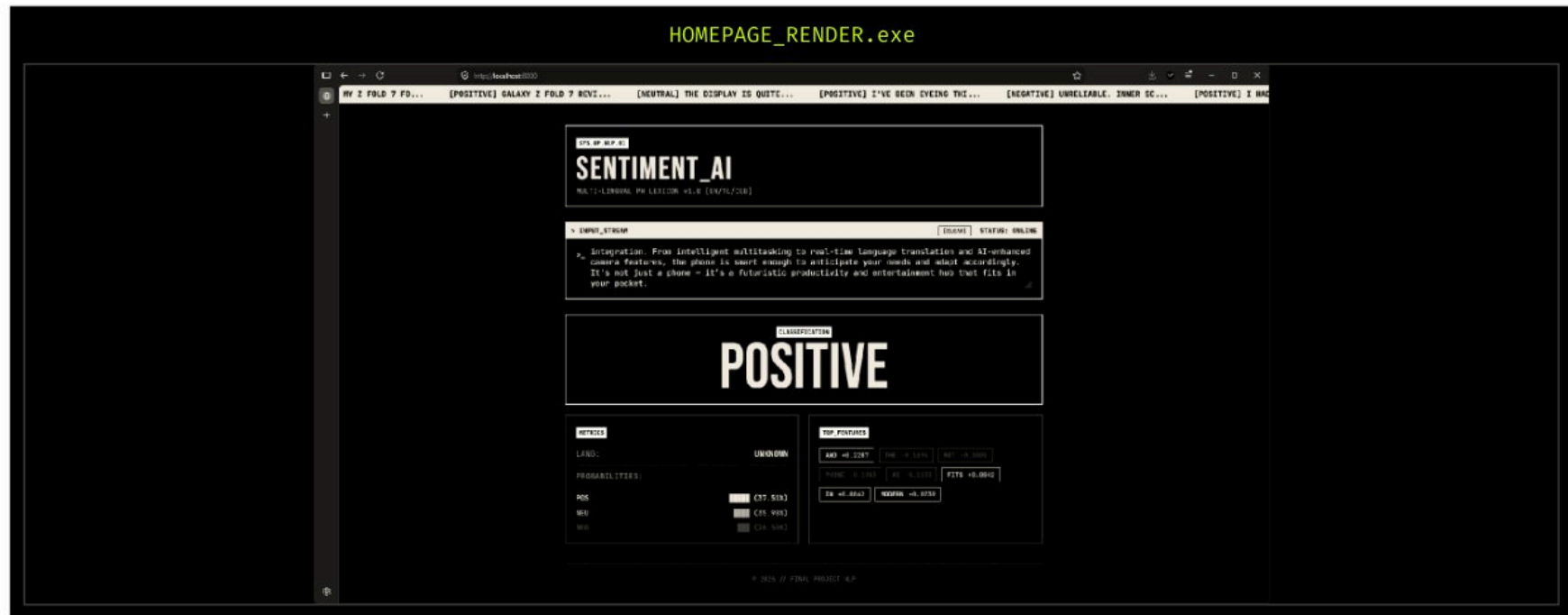
9. SYSTEM DEVELOPMENT TOOLS

- Python 3.10+ (Core logic)
- http.server (REST API Framework)
- Scikit-learn (ML Pipeline)
- Numpy & Pandas (Data manipulation)
- Emoji (Demojization)

- HTML5 / Vanilla JS
- Vanilla CSS
- Custom Acid Data Brutalism Design System
- Zero-bloat raw DOM rendering

10. SYSTEM DEMO (1/2)

> Acid Data Brutalism Landing | Zero-bloat layout | Raw text input terminal



11. SYSTEM DEMO (2/2)

- System tokenizes & intercepts emojis
- Pipeline infers sentiment
- UI renders real-time feature impact weights

```
{  
  "sentiment": "POSITIVE",  
  "confidence": 0.892,  
  "language": "english+tagalog",  
  "keywords": [  
    { "feature": "smiling_face_with_heart_eyes", "weight": 2.45 },  
    { "feature": "ganda", "weight": 1.82 }  
  ]  
}
```

12. RESULTS & EVALUATION

>85%

CLASSIFICATION ACCURACY

- Precision / Recall / F1-Score
- Per-language confusion matrices
- User satisfaction on XAI response times

```
// Confusion Matrix Sample (Tagalog)
{
  "True Positive": 1402,
  "False Positive": 128,
  "True Negative": 985,
  "False Negative": 150
}
```

13. CONCLUSION

- OBJECTIVES ACHIEVED: The system successfully implemented an emoji-aware pipeline, achieving >85% accuracy on code-mixed eCommerce data.
- MAJOR FINDINGS: Traditional ML coupled with XAI extraction is highly effective and computationally cheaper than Deep Learning. Emoji translation significantly boosts precision.
- CONTRIBUTION: Delivered a transparent, robust PH code-mixed NLP pipeline with an aggressive, dense visualization UI.

14. FUTURE ENHANCEMENTS

01

Annotate and integrate more human-checked Bisaya rows.

02

Benchmark traditional ML against deep learning BERT baselines.

03

Integrate multi-lingual spelling correction in the preprocessing pipeline.

04

Integrate Web Speech API for voice-to-text inference.

15. THE TEAM

#



Joseph Alejo

NLP DEVELOPER

Dataset collection
and preprocessing
pipelines.

Team members redacted

Team members redacted